

Concept and health-related properties of nonextractable polyphenols: the missing dietary polyphenols.



Most research studies in the field of dietary polyphenols or phenolic compounds use a chemical approach focusing exclusively on polyphenols extracted from plant foods with organic solvents. However, an appreciable part of polyphenols are not extracted with organic solvents and thus are ignored in biological, nutritional, and epidemiological studies. Recent studies have shown that these nonextractable polyphenols (NEPP) are a major part of total dietary polyphenols and that they exhibit a significant biological activity. A physiological approach is proposed on the basis that the bioavailability and health-related properties of polyphenols depend on their solubility in intestinal fluids, which is different from their solubility in organic solvents. This paper tries to clarify the concept of NEPP, distinguishing between chemical and physiological approaches and pointing out the main qualitative and quantitative differences between them. It is stressed that the literature and databases refer to only extractable polyphenols. Greater attention to NEPP may fill the current gap in the field of dietary polyphenols.